Ved Jaggi

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Education

Overseas Family School (OFS) Singapore
IGCSE & A-Levels *2020 — May 2026*

- Head of the OFS Mathematics Society — presented mathematical concepts to peers and wrote school-wide publications
- Leader of the Computer Programming Club — ran ML workshops and taught beginner Python
- Peer Mentor (3 years) and Peer Tutor (1 year, Peer Tutor Award) for IGCSE Additional & Extended Mathematics

Projects

Variational Autoencoder for Synthetic Data (vedjaggi.com) 2024

- Built a VAE in PyTorch to generate synthetic images of handwritten digits and fruit
- Implemented latent-space interpolation: encoding two images and walking the line between them produces smooth combinations
- Designed an interactive 2-D latent-space visualiser (Three.js) that maps 20-dimensional embeddings into a navigable scatter plot

p2pchat — Peer-to-Peer Encrypted Chat Network (github.com/DoggoofCode/p2pchat) 2025–2026

- Engineered a UDP-based P2P chat protocol with end-to-end RSA/AES encryption from scratch
- Designed a **ratification model**: every message is cryptographically approved by a designated ratifier node before broadcast, preventing forgery
- Implemented a Distributed Hash Table (DHT) for full message-history sync so any node joining the network immediately catches up
- Built the packet spec: message packets (create / edit / delete) and update packets (DHT sync, member list), documented in a public README

SSL — Simple Scripting Language (github.com/DoggoofCode/p2pchat) 2025–2026

- Designed and implemented an assembly-style interpreted language in Python with operator–operand syntax
- Built a 5-stage pipeline: resolver → lexer/tokeniser → label-tree builder → AST generator → executor
- Supports registers, named memory variables, arithmetic, conditional and unconditional jumps, nested label scoping, ANSI colour output, and blocking user input
- Deployed as an in-browser playground via Pyodide (WebAssembly) with ANSI rendering and inline terminal input

Portfolio Website (vedjaggi.com) 2026

- Designed and built a LaTeX/paper-styled interactive portfolio with Vite, Three.js, and marked.js
- Features: live SSL playground, VAE latent-space explorer, P2P chat network animation, MathJax typesetting, accordion folds, and CSS tooltips
- All content is authored in Markdown and rendered at runtime; deployed via Cloudflare Pages

Skills

- **Languages**: Python, JavaScript, TypeScript, HTML/CSS, SQL, Typst
- **ML / Data**: PyTorch, VAE, latent-space methods, data visualisation
- **Web**: Vite, Three.js, Pyodide (WASM), Canvas 2D, marked.js, MathJax
- **Systems**: UDP networking, custom binary/JSON protocols, cryptography (RSA, AES)
- **Tools**: Git, Cloudflare Pages, Linux/macOS CLI

Extracurricular Activities

Model United Nations (MUN) 2020 — Present

- Attended 25+ conferences since 2020 as delegate, Executive Team member, and Student Officer (STOFF)
- Represented school at TASMUN XV (international excursion, Taiwan)
- Coached peers on speech writing and ran mock debates for new MUN members